Chapter 24

Probability

Only One Option Correct Type Questions

1. Let E^c denote the complement of an event E. Let E, F, G be pairwise independent events with $P(G) > 0$ and $P(E \cap F \cap G) = 0$. Then $P(E^c \cap F^c \mid G)$ equals **[IIT-JEE-2007 (Paper-2)]**

   (A) $P(E^c) + P(F^c)$
   (B) $P(E^c) - P(F^c)$
   (C) $P(E^c) - P(F)$
   (D) $P(E) - P(F^c)$

2. An experiment has 10 equally likely outcomes. Let A and B be two non-empty events of the experiment. If A consists of 4 outcomes, the number of outcomes that B must have so that A and B are independent, is

   [IIT-JEE-2008 (Paper-2)]

   (A) 2, 4 or 8
   (B) 3, 6 or 9
   (C) 4 or 8
   (D) 5 or 10

3. Let ω be a complex cube root of unity with $\omega \neq 1$. A fair die is thrown three times. If r_1, r_2 and r_3 are the numbers obtained on the die, then the probability that $\omega^{r_1} + \omega^{r_2} + \omega^{r_3} = 0$ is **[IIT-JEE-2010 (Paper-1)]**

   (A) $\frac{1}{18}$
   (B) $\frac{1}{9}$
   (C) $\frac{2}{9}$
   (D) $\frac{1}{36}$

4. A signal which can be green or red with probability $\frac{4}{5}$ and $\frac{1}{5}$ respectively, is received by station A and then transmitted to station B. The probability of each station receiving the signal correctly is $\frac{3}{4}$. If the signal received at station B is green, then the probability that the original signal was green is

   [IIT-JEE-2010 (Paper-2)]

   (A) $\frac{3}{5}$
   (B) $\frac{6}{7}$
   (C) $\frac{20}{23}$
   (D) $\frac{9}{20}$

5. Four fair dice D_1, D_2, D_3 and D_4, each having six faces numbered 1, 2, 3, 4, 5 and 6, are rolled simultaneously. The probability that D_4 shows a number appearing on one of D_1, D_2 and D_3 is

   [IIT-JEE-2012 (Paper-2)]

   (A) $\frac{91}{216}$
   (B) $\frac{108}{216}$
   (C) $\frac{125}{216}$
   (D) $\frac{127}{216}$

6. Four persons independently solve a certain problem correctly with probabilities $\frac{1}{2}, \frac{3}{4}, \frac{1}{4}, \frac{1}{8}$. Then the probability that the problem is solved correctly by at least one of them is **[JEE (Adv)-2013 (Paper-1)]**

(A) $\frac{235}{256}$ (B) $\frac{21}{256}$

(C) $\frac{3}{256}$ (D) $\frac{253}{256}$

7. Three boys and two girls stand in a queue. The probability, that the number of boys ahead of every girl is at least one more than the number of girls ahead of her, is **[JEE (Adv)-2014 (Paper-2)]**

(A) $\frac{1}{2}$ (B) $\frac{1}{3}$

(C) $\frac{2}{3}$ (D) $\frac{3}{4}$

8. A computer producing factory has only two plants T_1 and T_2. Plant T_1 produces 20% and plant T_2 produces 80% of the total computers produced. 7% of computers produced in the factory turn out to be defective. It is known that P(computer turns out to be defective given that it is produced in plant T_1) = $10P$ (computer turns out to be defective given that it is produced in plant T_2) where $P(E)$ denotes the probability of an event E. A computer produced in the factory is randomly selected and it does not turn out to be defective. Then the probability that it is produced in plant T_2 is **[JEE (Adv)-2016 (Paper-1)]**

(A) $\frac{36}{73}$ (B) $\frac{47}{79}$

(C) $\frac{78}{93}$ (D) $\frac{75}{83}$

9. Three randomly chosen non-negative integers x, y and z are found to satisfy the equation $x + y + z = 10$. Then the probability that z is even, is **[JEE (Adv)-2017 (Paper-2)]**

(A) $\frac{6}{11}$ (B) $\frac{36}{55}$

(C) $\frac{1}{2}$ (D) $\frac{5}{11}$

10. Let C_1 and C_2 be two biased coins such that the probabilities of getting head in a single toss are $\frac{2}{3}$ and $\frac{1}{3}$, respectively. Suppose α is the number of heads that appear when C_1 is tossed twice, independently, and suppose β is the number of heads that appear when C_2 is tossed twice, independently. Then the probability that the roots of the quadratic polynomial $x^2 - \alpha x + \beta$ are real and equal, is **[JEE (Adv)-2020 (Paper-1)]**

(A) $\frac{40}{81}$ (B) $\frac{20}{81}$

(C) $\frac{1}{2}$ (D) $\frac{1}{4}$

11. Consider three sets $E_1 = \{1, 2, 3\}$, $F_1 = \{1, 3, 4\}$ and $G_1 = \{2, 3, 4, 5\}$. Two elements are chosen at random, without replacement, from the set E_1, and let S_1 denote the set of these chosen elements. Let $E_2 = E_1 - S_1$ and $F_2 = F_1 \cup S_1$. Now two elements are chosen at random, without replacement, from the set F_2 and let S_2 denote the set of these chosen elements.

Let $G_2 = G_1 \cup S_2$. Finally, two elements are chosen at random, without replacement from the set G_2 and let S_3 denote the set of these chosen elements.

Let $E_3 = E_2 \cup S_3$. Given that $E_1 = E_3$, let p be the conditional probability of the event $S_1 = \{1, 2\}$. Then the value of p is **[JEE (Adv)-2021 (Paper-1)]**

(A) $\frac{1}{5}$

(B) $\frac{3}{5}$

(C) $\frac{1}{2}$

(D) $\frac{2}{5}$

12. Suppose that **[JEE (Adv)-2022 (Paper-2)]**

Box-I contains 8 red, 3 blue and 5 green balls,
Box-II contains 24 red, 9 blue and 15 green balls,
Box-III contains 1 blue, 12 green and 3 yellow balls,
Box-IV contains 10 green, 16 orange and 6 white balls.

A ball is chosen randomly from Box-I; call this ball b. If b is red then a ball is chosen randomly from Box-II, if b is blue then a ball is chosen randomly from Box-III, and if b is green then a ball is chosen randomly from Box-IV. The conditional probability of the event ‘one of the chosen balls is white’ given that the event ‘at least one of the chosen balls is green’ has happened, is equal to

(A) $\frac{15}{256}$

(B) $\frac{3}{16}$

(C) $\frac{5}{52}$

(D) $\frac{1}{8}$

One or More Option(s) Correct Type Questions

13. Let E and F be two independent events. The probability that exactly one of them occurs is $\frac{11}{25}$ and the probability of none of them occurring is $\frac{2}{25}$. If $P(T)$ denotes the probability of occurrence of the event T, then **[IIT-JEE-2011 (Paper-2)]**

(A) $P(E) = \frac{4}{5}, P(F) = \frac{3}{5}$

(B) $P(E) = \frac{1}{5}, P(F) = \frac{2}{5}$

(C) $P(E) = \frac{2}{5}, P(F) = \frac{1}{5}$

(D) $P(E) = \frac{3}{5}, P(F) = \frac{4}{5}$

14. A ship is fitted with three engines E_1, E_2 and E_3. The engines function independently of each other with respective probabilities $\frac{1}{2}$, $\frac{1}{4}$ and $\frac{1}{4}$. For the ship to be operational at least two of its engines must function. Let X denote the event that the ship is operational and let X_1, X_2 and X_3 denote respectively the events that the engines E_1, E_2 and E_3 are functioning. Which of the following is (are) true? **[IIT-JEE-2012 (Paper-1)]**

(A) $P[X_1^c \mid X] = \frac{3}{16}$

(B) P[Exactly two engines of the ship are functioning $|X$] = $\frac{7}{8}$

(C) $P[X \mid X_2] = \frac{5}{16}$

(D) $P[X \mid X_1] = \frac{7}{16}$

15. Let X and Y be two events such that $P(X \mid Y) = \frac{1}{2}$, $P(Y \mid X) = \frac{1}{3}$ and $P(X \cap Y) = \frac{1}{6}$. Which of the following is (are) correct? **[IIT-JEE-2012 (Paper-2)]**

(A) $P(X \cup Y) = \frac{2}{3}$

(B) X and Y are independent

(C) X and Y are not independent

(D) $P(X^c \cap Y) = \frac{1}{3}$

16. Let X and Y be two events such that $P(X) = \frac{1}{3}$, $P(X \mid Y) = \frac{1}{2}$ and $P(Y \mid X) = \frac{2}{5}$. Then

[JEE (Adv)-2017 (Paper-1)]

(A) $P(X \cap Y) = \frac{1}{5}$

(B) $P(Y) = \frac{4}{15}$

(C) $P(X' \mid Y) = \frac{1}{2}$

(D) $P(X \cup Y) = \frac{2}{5}$

17. There are three bags B_1, B_2 and B_3. The bag B_1 contains 5 red and 5 green balls, B_2 contains 3 red and 5 green balls and B_3 contains 5 red and 3 green balls. Bags B_1, B_2 and B_3 have probabilities $\frac{3}{10}$, $\frac{3}{10}$ and $\frac{4}{10}$ respectively of being chosen. A bag is selected at random and a ball is chosen at random from the bag. Then which of the following options is/are correct? **[JEE (Adv)-2019 (Paper-1)]**

(A) Probability that the selected bag is B_3 and the chosen ball is green equals $\frac{3}{10}$

(B) Probability that the selected bag is B_3, given that the chosen ball is green, equals $\frac{5}{13}$

(C) Probability that the chosen ball is green, given that the selected bag is B_3, equals $\frac{3}{8}$

(D) Probability that the chosen ball is green equals $\frac{39}{80}$

18. Let E, F and G be three events having probabilities

$$P(E)=\frac{1}{8}, P(F)=\frac{1}{6} \text{ and } P(G)=\frac{1}{4}, \text{ and } P(E\cap F\cap G)=\frac{1}{10}.$$

For any event H, if H^c denotes its complement, then which of the following statements is(are) **TRUE**?

[JEE (Adv)-2021 (Paper-1)]

(A) $P\left(E\cap F\cap G^c\right)\le\frac{1}{40}$

(B) $P\left(E^c\cap F\cap G\right)\le\frac{1}{15}$

(C) $P\left(E\cup F\cup G\right)\le\frac{13}{24}$

(D) $P\left(E^c\cap F^c\cap G^c\right)\le\frac{5}{12}$

Linked Comprehension Type Questions

Paragraph for Q.Nos. 19 to 21

A fair die is tossed repeatedly until a six is obtained. Let X denote the number is tosses required.

[IIT-JEE-2009 (Paper-1)]

Choose the correct answer :

19. The probability that $X = 3$, equals

(A) $\frac{25}{216}$

(B) $\frac{25}{36}$

(C) $\frac{5}{36}$

(D) $\frac{125}{216}$

20. The probability that $X \ge 3$ equals

(A) $\frac{125}{216}$

(B) $\frac{25}{36}$

(C) $\frac{5}{36}$

(D) $\frac{25}{216}$

21. The conditional probability that $X \ge 6$ given $X = 3$ equals

(A) $\frac{125}{216}$

(B) $\frac{25}{216}$

(C) $\frac{5}{36}$

(D) $\frac{25}{36}$

Paragraph for Q.Nos. 22 and 23

Let U_1 and U_2 be two urns such that U_1 contains 3 white and 2 red balls, and U_2 contains only 1 white ball. A fair coin is tossed. If head appears then 1 ball is drawn at random from U_1 and put into U_2. However, if tail appears then 2 balls are drawn at random from U_1 and put into U_2. Now 1 ball is drawn at random from U_2.

[IIT-JEE-2011 (Paper-1)]

Choose the correct answer :

22. The probability of the drawn ball from U_2 being white is

(A) $\frac{13}{30}$ (B) $\frac{23}{30}$

(C) $\frac{19}{30}$ (D) $\frac{11}{30}$

23. Given that the drawn ball from U_2 is white, the probability that head appeared on the coin is

(A) $\frac{17}{23}$ (B) $\frac{11}{23}$

(C) $\frac{15}{23}$ (D) $\frac{12}{23}$

Paragraph for Q.Nos. 24 and 25

A box B_1 contains 1 white ball, 3 red balls and 2 black balls. Another box B_2 contains 2 white balls, 3 red balls and 4 black balls. A third box B_3 contains 3 white balls, 4 red balls and 5 black balls.

[JEE (Adv)-2013 (Paper-2)]

Choose the correct answer :

24. If 1 ball is drawn from each of the boxes B_1, B_2 and B_3, the probability that all 3 drawn balls are of the same colour is

(A) $\frac{82}{648}$ (B) $\frac{90}{648}$

(C) $\frac{558}{648}$ (D) $\frac{566}{648}$

25. If 2 balls are drawn (without replacement) from a randomly selected box and one of the balls is white and the other ball is red, the probability that these 2 balls are drawn from box B_2 is

(A) $\frac{116}{181}$ (B) $\frac{126}{181}$

(C) $\frac{65}{181}$ (D) $\frac{55}{181}$

Paragraph for Q.Nos. 26 and 27

Box 1 contains three cards bearing numbers 1, 2, 3; box 2 contains five cards bearing numbers 1, 2, 3, 4, 5; and box 3 contains seven cards bearing numbers 1, 2, 3, 4, 5, 6, 7. A card is drawn from each of the boxes. Let x_i be number on the card drawn from the i^{th} box, i = 1, 2, 3. **[JEE (Adv)-2014 (Paper-2)]**

Choose the correct answer :

26. The probability that $x_1 + x_2 + x_3$ is odd, is

(A) $\frac{29}{105}$ (B) $\frac{53}{105}$

(C) $\frac{57}{105}$ (D) $\frac{1}{2}$

27. The probability that x_1, x_2, x_3 are in an arithmetic progression, is

(A) $\frac{9}{105}$ (B) $\frac{10}{105}$

(C) $\frac{11}{105}$ (D) $\frac{7}{105}$

Paragraph for Q.Nos. 28 and 29

Let n_1 and n_2 be the number of red and black balls, respectively, in box I. Let n_3 and n_4 be the number of red and black balls, respectively, in box II. **[JEE (Adv)-2015 (Paper-2)]**

Choose the correct answer :

28. One of the two boxes, box I and box II, was selected at random and a ball was drawn randomly out of this box. The ball was found to be red. If the probability that this red ball was drawn from box II is $\frac{1}{3}$, then the correct option(s) with the possible values of n_1, n_2, n_3 and n_4 is(are)

(A) $n_1 = 3$, $n_2 = 3$, $n_3 = 5$, $n_4 = 15$

(B) $n_1 = 3$, $n_2 = 6$, $n_3 = 10$, $n_4 = 50$

(C) $n_1 = 8$, $n_2 = 6$, $n_3 = 5$, $n_4 = 20$

(D) $n_1 = 6$, $n_2 = 12$, $n_3 = 5$, $n_4 = 20$

29. A ball is drawn at random from box I and transferred to box II. If the probability of drawing a red ball from box I, after this transfer, is $\frac{1}{3}$, then the correct option(s) with the possible values of n_1 and n_2 is(are)

(A) $n_1 = 4$ and $n_2 = 6$

(B) $n_1 = 2$ and $n_2 = 3$

(C) $n_1 = 10$ and $n_2 = 20$

(D) $n_1 = 3$ and $n_2 = 6$

Paragraph for Q.Nos. 30 and 31

Football teams T_1 and T_2 have to play two games against each other. It is assumed that the outcomes of the two games are independent. The probabilities of T_1 winning, drawing and losing a game against T_2 are $\frac{1}{2}, \frac{1}{6}$ and $\frac{1}{3}$, respectively. Each team gets 3 points for a win, 1 point for a draw and 0 point for a loss in a game. Let X and Y denote the total points scored by teams T_1 and T_2, respectively, after two games.

[JEE (Adv)-2016 (Paper-2)]

Choose the correct answer :

30. $P(X > Y)$ is

(A) $\frac{1}{4}$ (B) $\frac{5}{12}$

(C) $\frac{1}{2}$ (D) $\frac{7}{12}$

31. $P(X = Y)$ is

(A) $\frac{11}{36}$ (B) $\frac{1}{3}$

(C) $\frac{13}{36}$ (D) $\frac{1}{2}$

Paragraph for Q.Nos. 32 and 33

There are five students S_1, S_2, S_3, S_4 and S_5 in a music class and for them there are five seats R_1, R_2, R_3, R_4 and R_5 arranged in a row, where initially the seat R_i is allotted to the student S_i, i = 1, 2, 3, 4, 5. But, on the examination day, the five students are randomly allotted the five seats. **[JEE (Adv)-2018 (Paper-1)]**

Choose the correct answer :

31. The probability that, on the examination day, the student S_1 gets the previously allotted seat R_1, and **NONE** of the remaining students gets the seat previously allotted to him/her is

(A) $\frac{3}{40}$

(B) $\frac{1}{8}$

(C) $\frac{7}{40}$

(D) $\frac{1}{5}$

32. For i = 1, 2, 3, 4, let T_i denote the event that the students S_i and S_{i+1} do **NOT** sit adjacent to each other on the day of the examination. Then, the probability of the event $T_1 \cap T_2 \cap T_3 \cap T_4$ is

(A) $\frac{1}{15}$

(B) $\frac{1}{10}$

(C) $\frac{7}{60}$

(D) $\frac{1}{5}$

Integer / Numerical Value Type Questions

33. Of the three independent events E_1, E_2 and E_3, the probability that only E_1 occurs is α, only E_2 occurs is β and only E_3 occurs is γ. Let the probability p that none of events E_1, E_2 or E_3 occurs satisfy the equations $(\alpha - 2\beta)p = \alpha\beta$ and $(\beta - 3\gamma)p = 2\beta\gamma$. All the given probabilities are assumed to lie in the interval (0, 1). Then

$$\frac{\text{Probability of occurrence of } E_1}{\text{Probability of occurrence of } E_3} =$$ **[JEE (Adv)-2013 (Paper-1)]**

34. The minimum number of times a fair coin needs to be tossed, so that the probability of getting at least two heads is at least 0.96, is **[JEE (Adv)-2015 (Paper-1)]**

35. Let S be the sample space of all 3 × 3 matrices with entries from the set {0, 1}. Let the events E_1 and E_2 be given by

$E_1 = \{A \in S : \det A = 0\}$ and

$E_2 = \{A \in S : \text{sum of entries of } A \text{ is } 7\}$.

If a matrix is chosen at random from S, then the conditional probability $P(E_1|E_2)$ equals ___.

[JEE (Adv)-2019 (Paper-1)]

36. Let $|X|$ denote the number of elements in a set X. Let $S = \{1, 2, 3, 4, 5, 6\}$ be a sample space, where each element is equally likely to occur. If A and B are independent events associated with S, then the number of ordered pairs (A, B) such that $1 \le |B| < |A|$, equals ____ **[JEE (Adv)-2019 (Paper-2)]**

37. The probability that a missile hits a target successfully is 0.75. In order to destroy the target completely, at least three successful hits are required. Then the minimum number of missiles that have to be fired so that the probability of completely destroying the target is **NOT** less than 0.95, is _____

[JEE (Adv)-2020 (Paper-2)]

38. Two fair dice, each with faces numbered 1,2,3,4,5 and 6, are rolled together and the sum of the numbers on the faces is observed. This process is repeated till the sum is either a prime number or a perfect square. Suppose the sum turns out to be a perfect square before it turns out to be a prime number. If p is the probability that this perfect square is an odd number, then the value of $14p$ is _____

[JEE (Adv)-2020 (Paper-2)]

Question Stem for Question Nos. 39 and 40

Question Stem

Three numbers are chosen at random, one after another with replacement, from the set $S = \{1, 2, 3, ..., 100\}$. Let p_1 be the probability that the maximum of chosen numbers is at least 81 and p_2 be the probability that the minimum of chosen numbers is at most 40. **[JEE (Adv)-2021 (Paper-1)]**

39. The value of $\frac{625}{4}p_1$ is _____.

40. The value of $\frac{125}{4}p_2$ is _____.

41. A number is chosen at random from the set {1, 2, 3......, 2000}. Let p be the probability that the number is a multiple of 3 or a multiple of 7. Then the value of $500p$ is ______. **[JEE (Adv)-2021 (Paper-2)]**

42. In a study about a pandemic, data of 900 persons was collected. It was found that **[JEE (Adv)-2022 (Paper-1)]**

190 persons had symptom of fever,

220 persons had symptom of cough,

220 persons had symptom of breathing problem,

330 persons had symptom of fever or cough or both,

350 persons had symptom of cough or breathing problem or both,

340 persons had symptom of fever or breathing problem or both,

30 persons had all three symptoms (fever, cough and breathing problem).

If a person is chosen randomly from these 900 persons, then the probability that the person has at most one symptom is _____________.

Matrix-Match / Matrix Based Type Questions

43. Two players, P_1 and P_2, play a game against each other. In every round of the game, each player rolls a fair die once, where the six faces of the die have six distinct numbers. Let x and y denote the readings on the die rolled by P_1 and P_2, respectively. If $x > y$, then P_1 scores 5 points and P_2 scores 0 point. If $x = y$, then each player scores 2 points. If $x < y$, then P_1 scores 0 point and P_2 scores 5 points. Let X_i and Y_i be the total scores of P_1 and P_2, respectively, after playing the i^{th} round. **[JEE (Adv)-2022 (Paper-1)]**

List-I	**List-II**
(I) Probability of $(X_2 \geq Y_2)$ is	(P) $\frac{3}{8}$
(II) Probability of $(X_2 > Y_2)$ is	(Q) $\frac{11}{16}$
(III) Probability of $(X_3 = Y_3)$ is	(R) $\frac{5}{16}$
(IV) Probability of $(X_3 > Y_3)$ is	(S) $\frac{355}{864}$
	(T) $\frac{77}{432}$

The correct option is:

(A) (I) $\rightarrow$ (Q); (II) $\rightarrow$ (R); (III) $\rightarrow$ (T); (IV) $\rightarrow$ (S)

(B) (I) $\rightarrow$ (Q); (II) $\rightarrow$ (R); (III) $\rightarrow$ (T); (IV) $\rightarrow$ (T)

(C) (I) $\rightarrow$ (P); (II) $\rightarrow$ (R); (III) $\rightarrow$ (Q); (IV) $\rightarrow$ (S)

(D) (I) $\rightarrow$ (P); (II) $\rightarrow$ (R); (III) $\rightarrow$ (Q); (IV) $\rightarrow$ (T)

❑ ❑ ❑